Microscopic Image Deblurring by a Generative Adversarial Network for 2D Nanomaterials: Implications for Wafer-Scale Semiconductor Characterization

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Wafer-scale two-dimensional (2D) semiconductors with atomically thin layers are promising materials for fabricating optic and photonic devices. Bright-field microscopy is one of the most efficient methods for large-area characterization and quality control. However, a significant challenge is the focus drift that occurs during long scanning processes, which results in blurred images and limits the accuracy of automated analysis.To address this, we propose a deep learning approach to restore these out-of-focus images. We developed a Generative Adversarial Network (GAN) using a combined loss function (MS-SSIM and L1) to effectively recover both structural details and color fidelity. Our results show that the model achieves a high structural similarity index (SSIM $>$ 90%) compared to the ground truth. Furthermore, by using these restored images for segmentation, we were able to automatically identify monolayer and multilayer regions with over 80% accuracy. This work demonstrates that machine learning can be a practical solution to overcome hardware limitations in the microscopic characterization of 2D nanomaterials.